Artificial Intelligence Concepts and Machine Learning Techniques (AI – 103)

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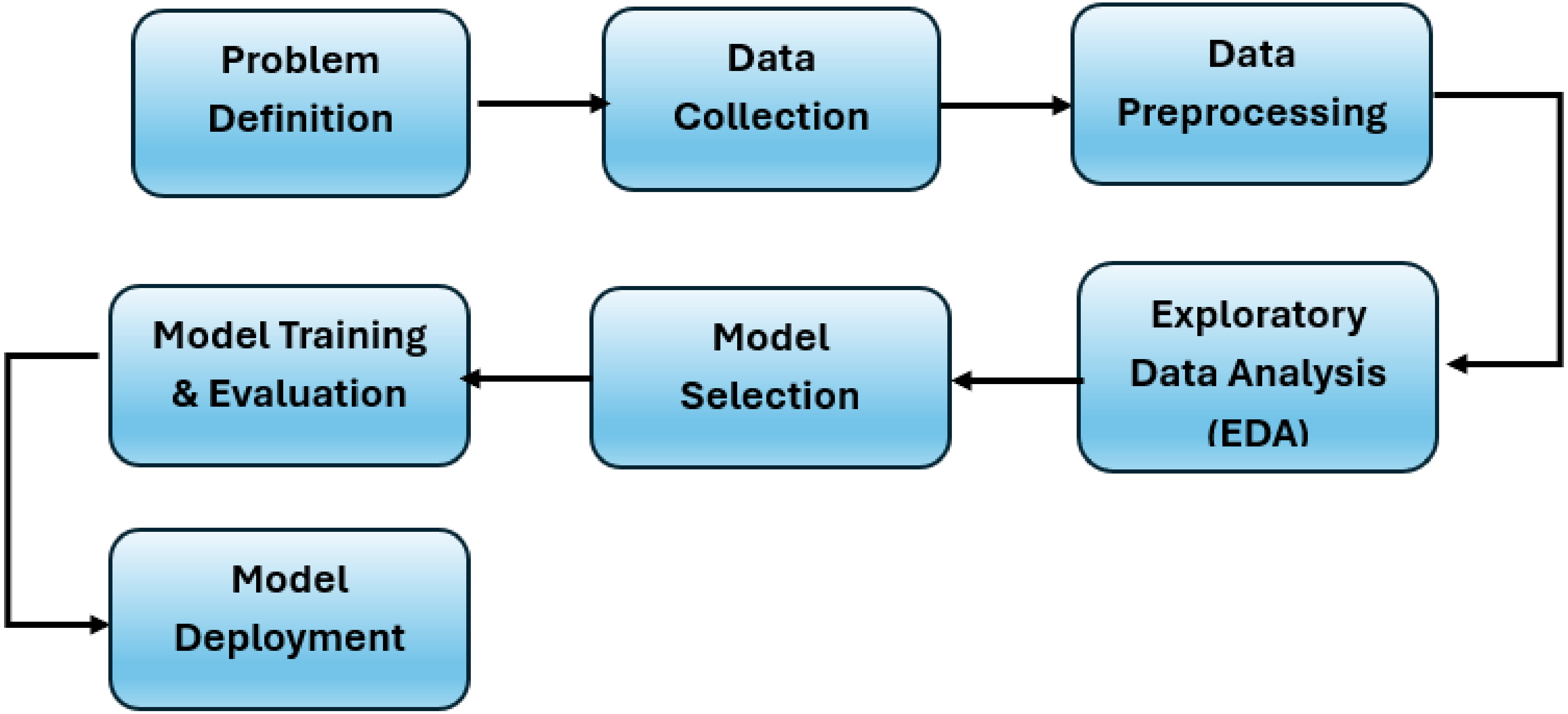
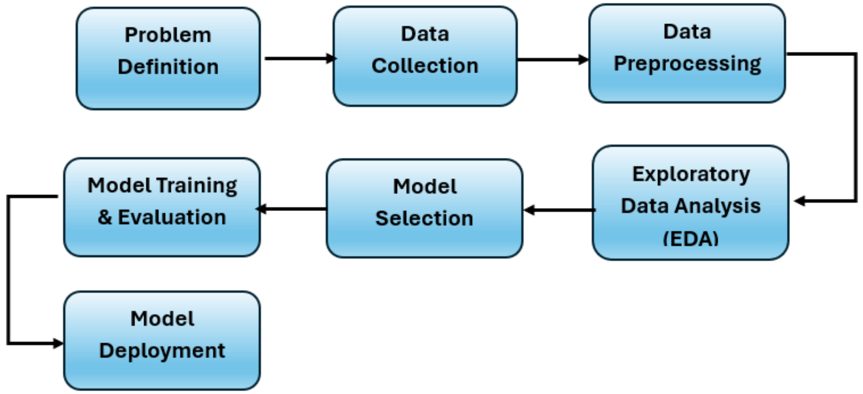
Academic Profiles: Google Scholar Scopus ResearchGate LinkedIn

Machine Learning workflow and pipelines


Problem Definition
Data Collection
Data Preprocessing
Exploratory Data Analysis (EDA)
Model Selection
Model Training & Evaluation
Model Deployment

Phases:



Problem definition/understanding:

- Clearly understand the business problem or the objective that needs to be addressed with ML.
- Define the scope and constraints of the project, including available resources, data, and time.

Data collection:

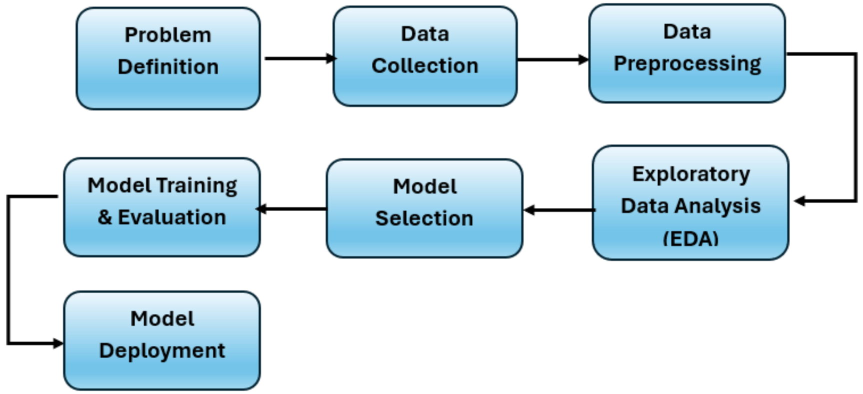
- Identify and gather relevant data from various sources such as databases, APIs, web scraping, sensors, etc.
- Ensure data is collected in a structured and organized manner.

Data preprocessing (cleaning and preparation):

- Check for missing data, duplicates, and outliers, and handle them appropriately.
- Transform and format the data to make it suitable for analysis.

most important step in the entire life cycle

Phases:



Exploratory Data analysis:

- Explore the data using summary statistics and data visualization techniques.
- Identify patterns, correlations, and trends to gain initial insights into the data.

getting some idea about the solution and factors affecting it before building the actual model

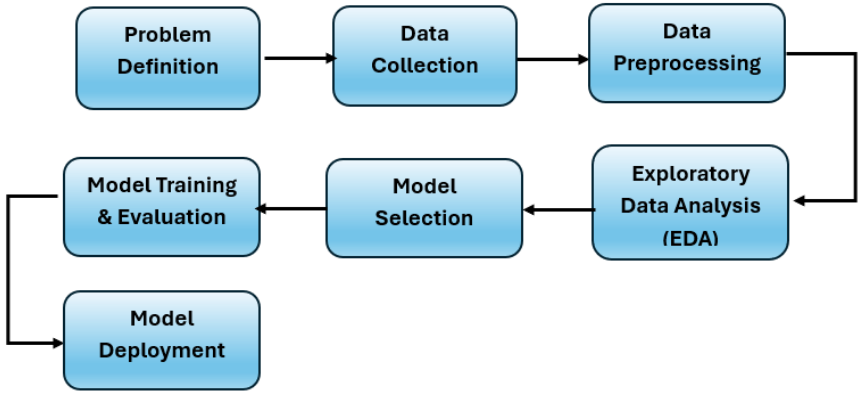
Feature engineering:

- Create new features or derive useful information from existing features to enhance model performance.

Model selection:

- Select appropriate machine learning algorithms based on the problem and data characteristics.
- Split the data into training and testing sets to evaluate the model's performance.

Phases:

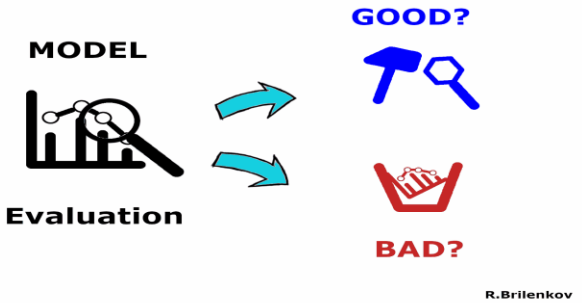


Model training:

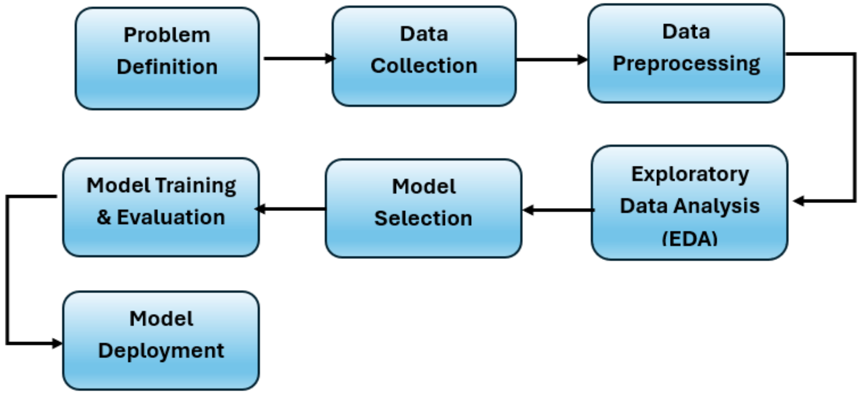
- Fit the selected models to the training data whether the problem is a classification problem, or a regression problem or a clustering problem.
- Optimize model parameters and hyperparameters using techniques like cross-validation and grid search.

Model evaluation:

- Evaluate the trained models using appropriate evaluation metrics (e.g., accuracy, precision, recall, F1-score, mean squared error, etc.)
- Compare different models to select the best- performing one.



Phases:



Data visualization and model deployment:

- Present the findings, insights, and predictions to stakeholders in a clear and understandable manner.
- Use data visualization tools to create interactive and informative visualizations.
- Deploy the chosen model into the production environment to make predictions on new, unseen data.

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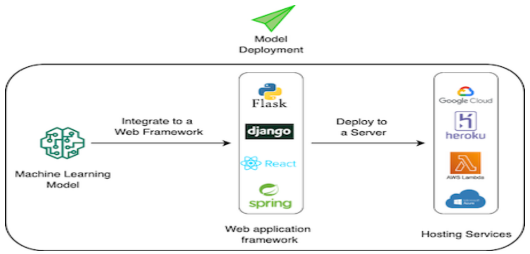
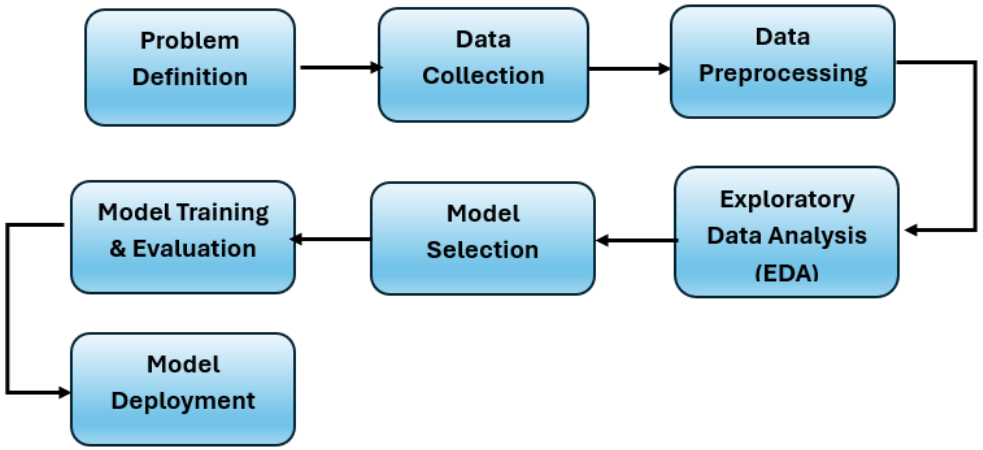


Monitor and Maintain:

- Continuously monitor the model's performance in the production environment.
- Make necessary updates or improvements as the data and requirements change.

Workflow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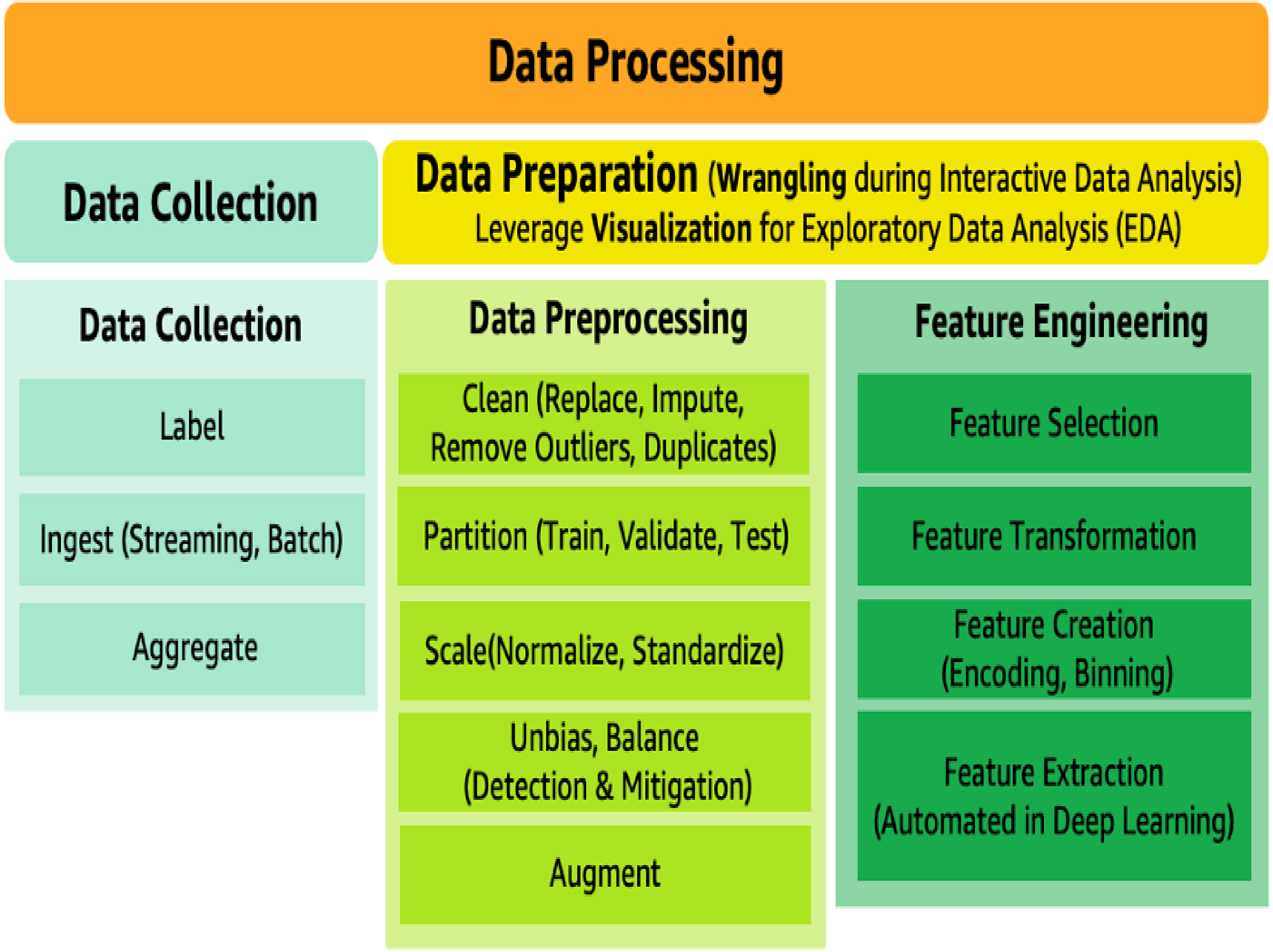

Data Processing
Data Collection
Data Preparation (Wrangling during Interactive Data Analysis)
Leverage Visualization for Exploratory Data Analysis (EDA)
Data Collection
Label
Ingest (Streaming, Batch)
Aggregate
Data Preprocessing
Clean (Replace, Impute, Remove Outliers, Duplicates)
Partition (Train, Validate, Test)
Scale(Normalize, Standardize)
Unbias, Balance (Detection & Mitigation)
Augment
Feature Engineering
Feature Selection
Feature Transformation
Feature Creation (Encoding, Binning)
Feature Extraction (Automated in Deep Learning)

Data collection

Data collection is the process of obtaining raw data from various sources, such as databases, APIs, files, websites, or sensors.

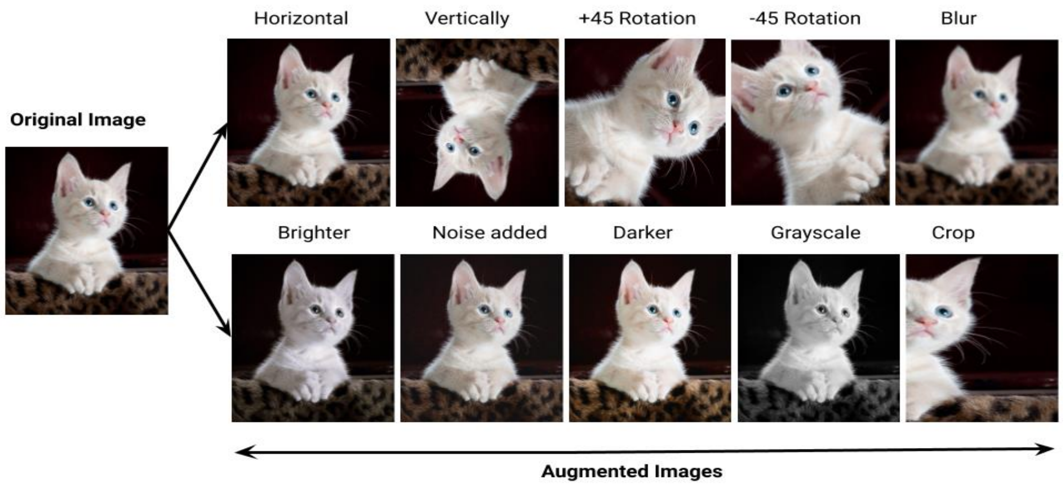
- The data can be structured (e.g., in a tabular format like CSV, Excel, or SQL databases) or unstructured (e.g., text, images, audio, video). The data acquired should be relevant and representative of the problem at hand.

- The primary goal of data acquisition is to collect sufficient and meaningful data to train the machine learning model effectively.

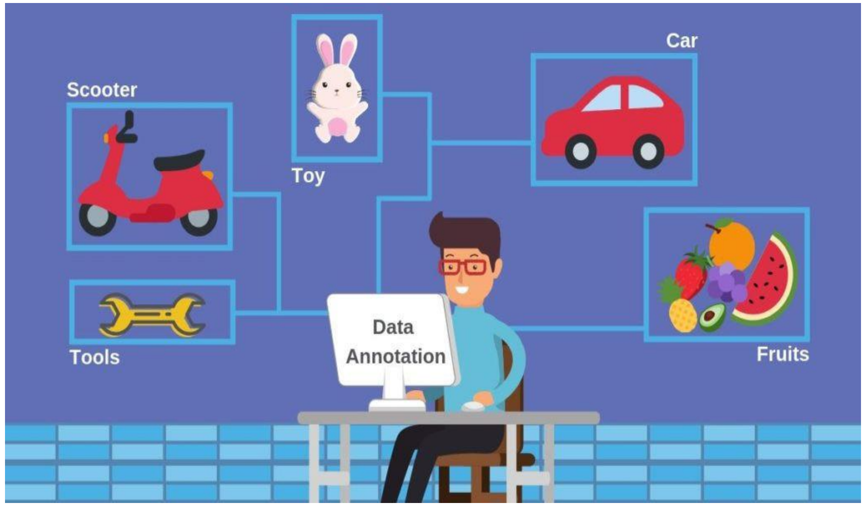
This step may include Data augmentation and Data labeling

Data collection

Data augmentation: In some cases, data augmentation might be required to expand the size of the existing dataset without gathering more data. For example, if a dataset of images were collected, they can be augmented by rotating the original versions, cropping them differently, or altering the lighting conditions.



Data labeling/Annotation: it involves processing unlabeled data like images, videos, text, and audio by adding one or more meaningful labels to provide context for the models so they can use the labeled data to make accurate predictions.



Data preparation

Data preparation is the process of constructing dataset correctly and transforming raw data into an efficient and useful format.

Data pre-processing involves cleaning and preparing raw data to facilitate feature engineering.

Data pre-processing primarily focuses on cleaning and organizing the data.

Feature engineering entails employing various techniques to manipulate the data. This may include adding or removing relevant features, handling missing data, encoding variables, and dealing with categorical variables, among other tasks.

Feature engineering is a critical task that significantly influences the outcome of a model. It involves crafting new features based on existing data.

Data preprocessing

Data Collection/Acquisition and Preprocessing are essential steps in the machine learning workflow.

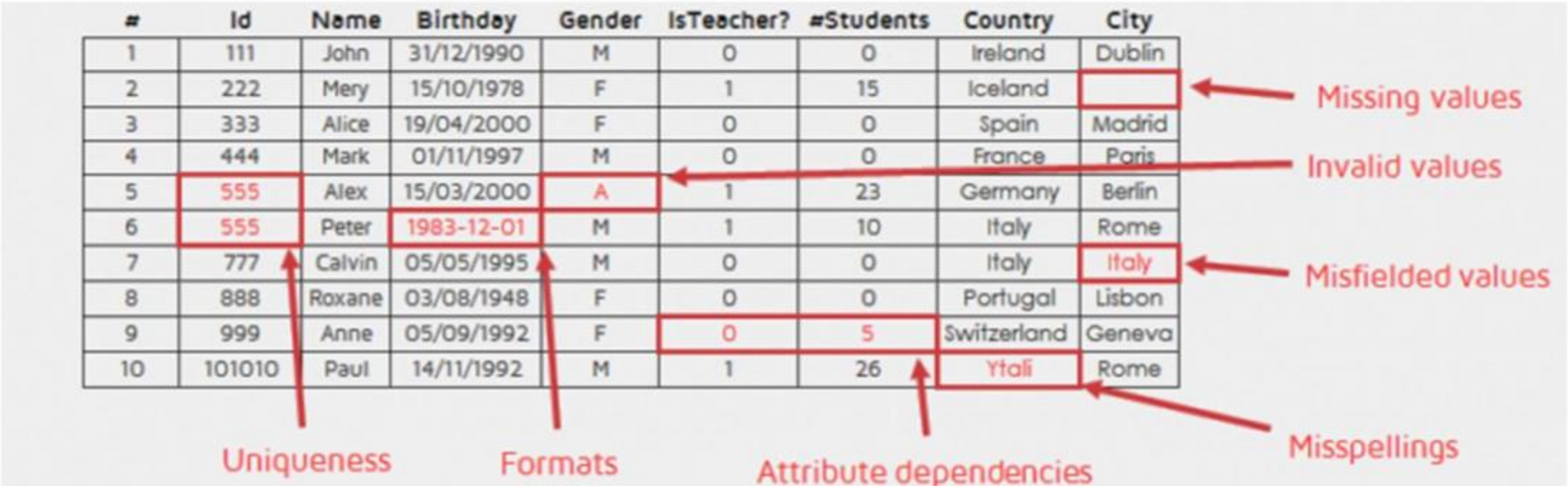
Data Preprocessing: The process of transforming the raw data into a format suitable for machine learning algorithms.

- This step is essential because real-world data is often noisy, incomplete, or inconsistent, and machine learning models require clean and standardized data to learn patterns effectively.

This step may include Data cleaning, handling outliers, splitting data and more

Key steps involved in data preprocessing

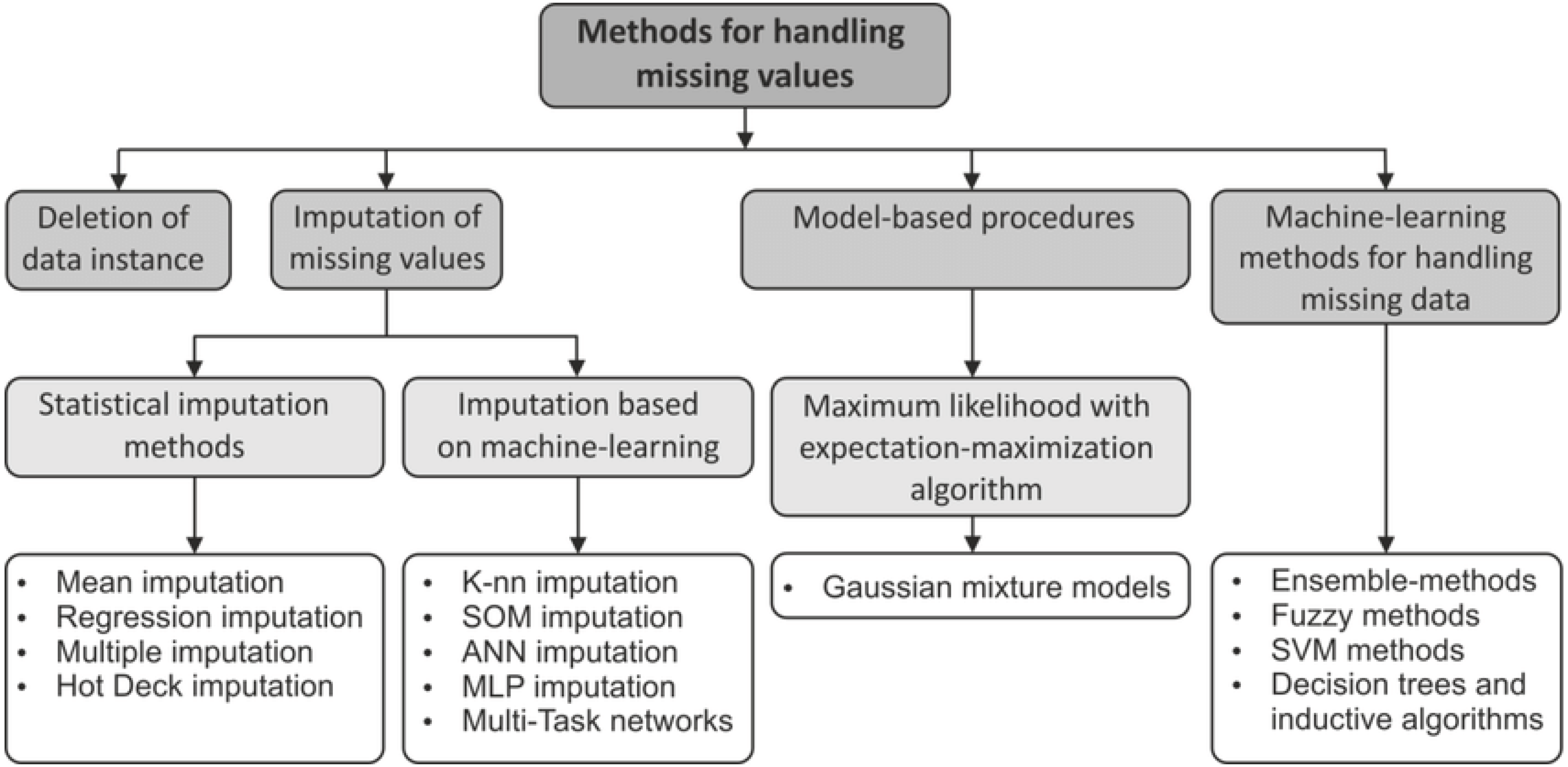
- **Data Cleaning**: *This step deals with handling missing values, removing duplicates, and correcting any inaccuracies in the data.*



#	Id	Name	Birthday	Gender	IsTeacher?	#Students	Country	City
1	111	John	31/12/1990	M	0	0	Ireland	Dublin
2	222	Mery	15/10/1978	F	1	15	Iceland	
3	333	Alice	19/04/2000	F	0	0	Spain	Madrid
4	444	Mark	01/11/1997	M	0	0	France	Paris
5	555	Alex	15/03/2000	A	1	23	Germany	Berlin
6	555	Peter	1983-12-01	M	1	10	Italy	Rome
7	777	Calvin	05/05/1995	M	0	0	Italy	Italy
8	888	Roxane	03/08/1948	F	0	0	Portugal	Lisbon
9	999	Anne	05/09/1992	F	0	5	Switzerland	Geneva
10	101010	Paul	14/11/1992	M	1	26	Ytali	Rome

Key steps involved in data preprocessing

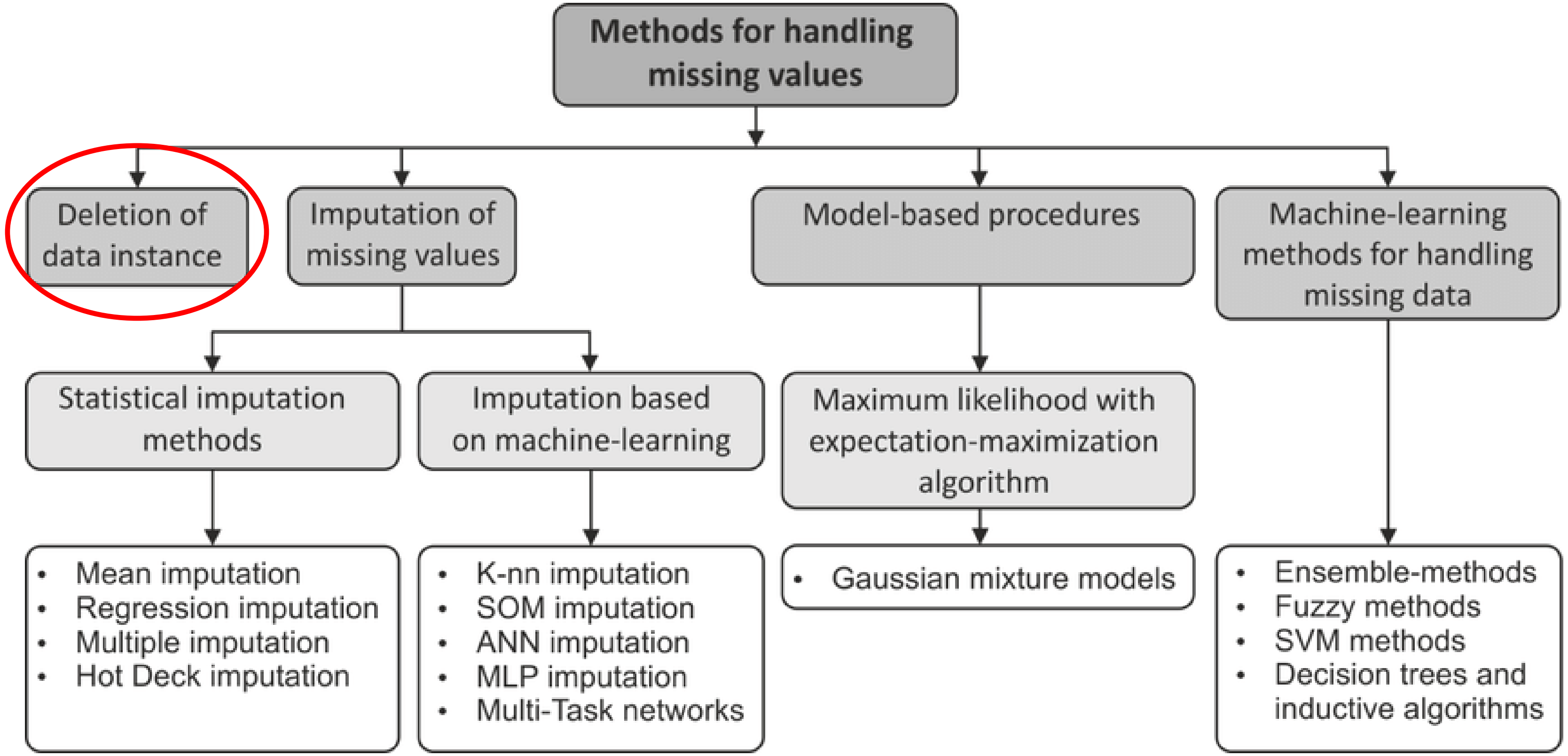
- **Handling missing values**: *it is a critical step in data preprocessing in which many machine learning algorithms can not handle missing data directly. The choice of handling technique depends on the nature of data and the problem at hand. Missing data can also refer to as NA(Not Available) or NaN(Not A Number).*



Deletion of data instance

- If the dataset contains a relatively small number of missing values, you can simply remove the rows that have missing values. However, this approach should be used with caution, as it can lead to a significant loss of data.

- It can be used if the missing values less than 5% of the total available data.

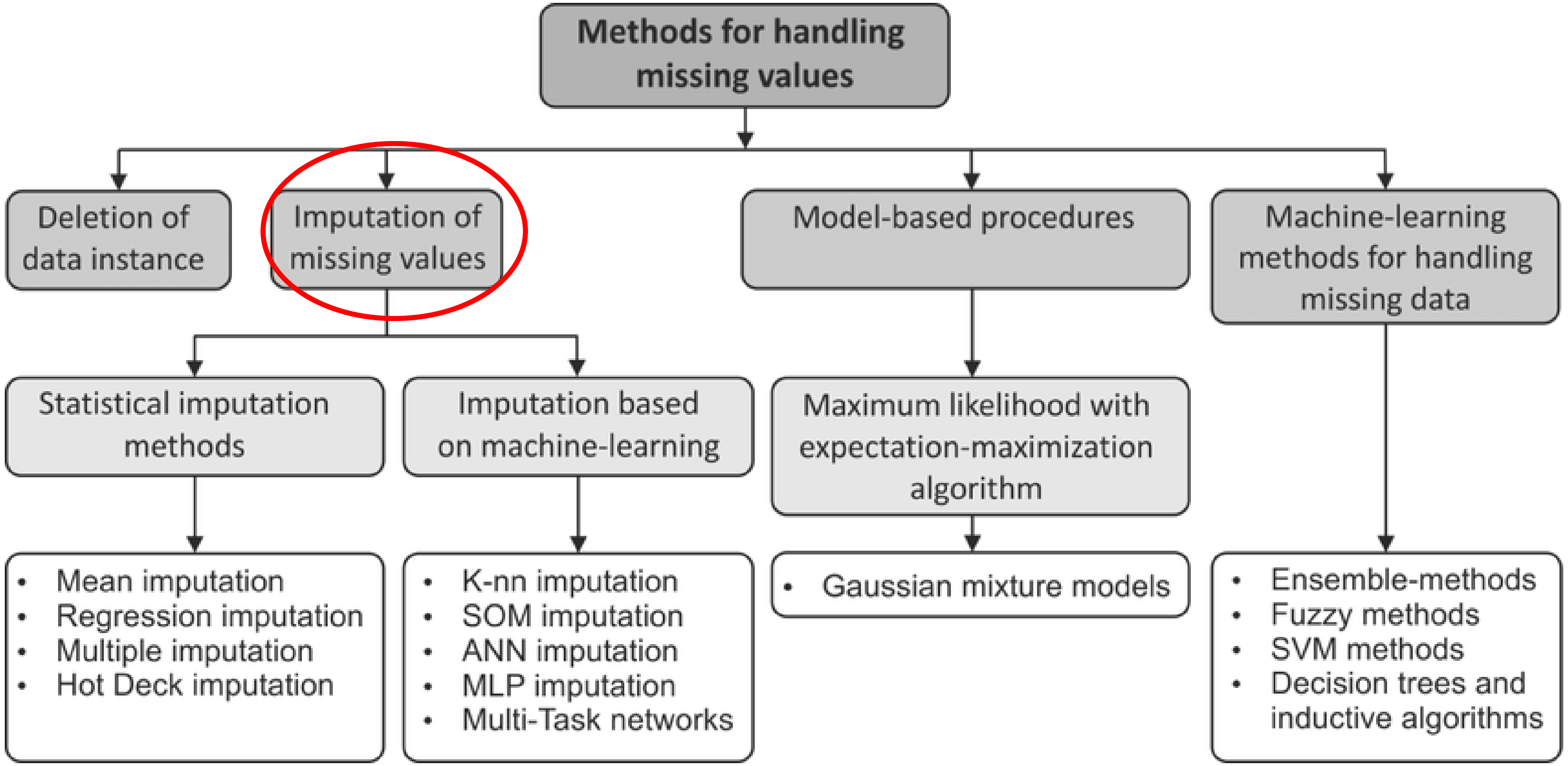


Imputation of missing values

- Predict the most appropriate value to replace missing values using different statistical methods/ML.

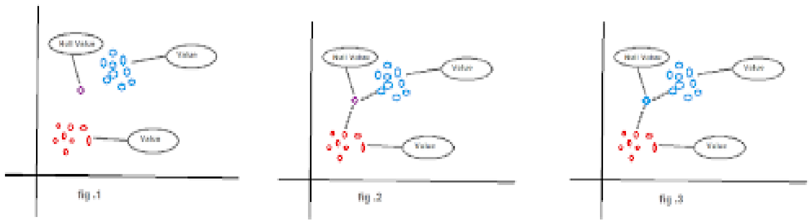
- Missing Values are calculated with different methods based on the data type of that specific variable.

- Mean and median for numerical data with a normal distribution while mode for categorical data, replace missing values with the mode (most frequent value) of the column.



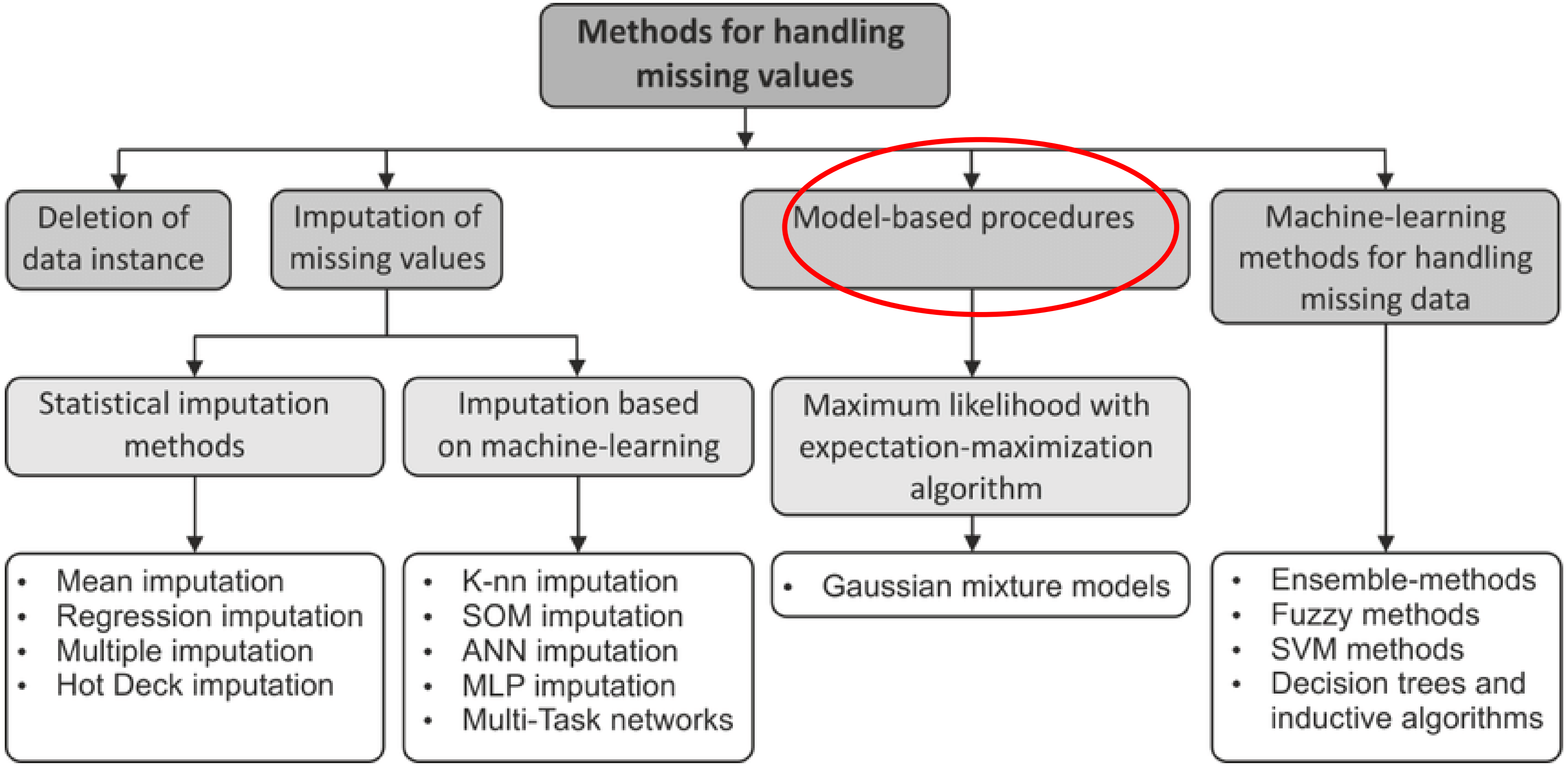
For example:

KNN works by finding the distances between a query and all the examples in the data, If NaN Values are found then it can be replaced with the nearest neighbour estimated values.



Model-based procedures

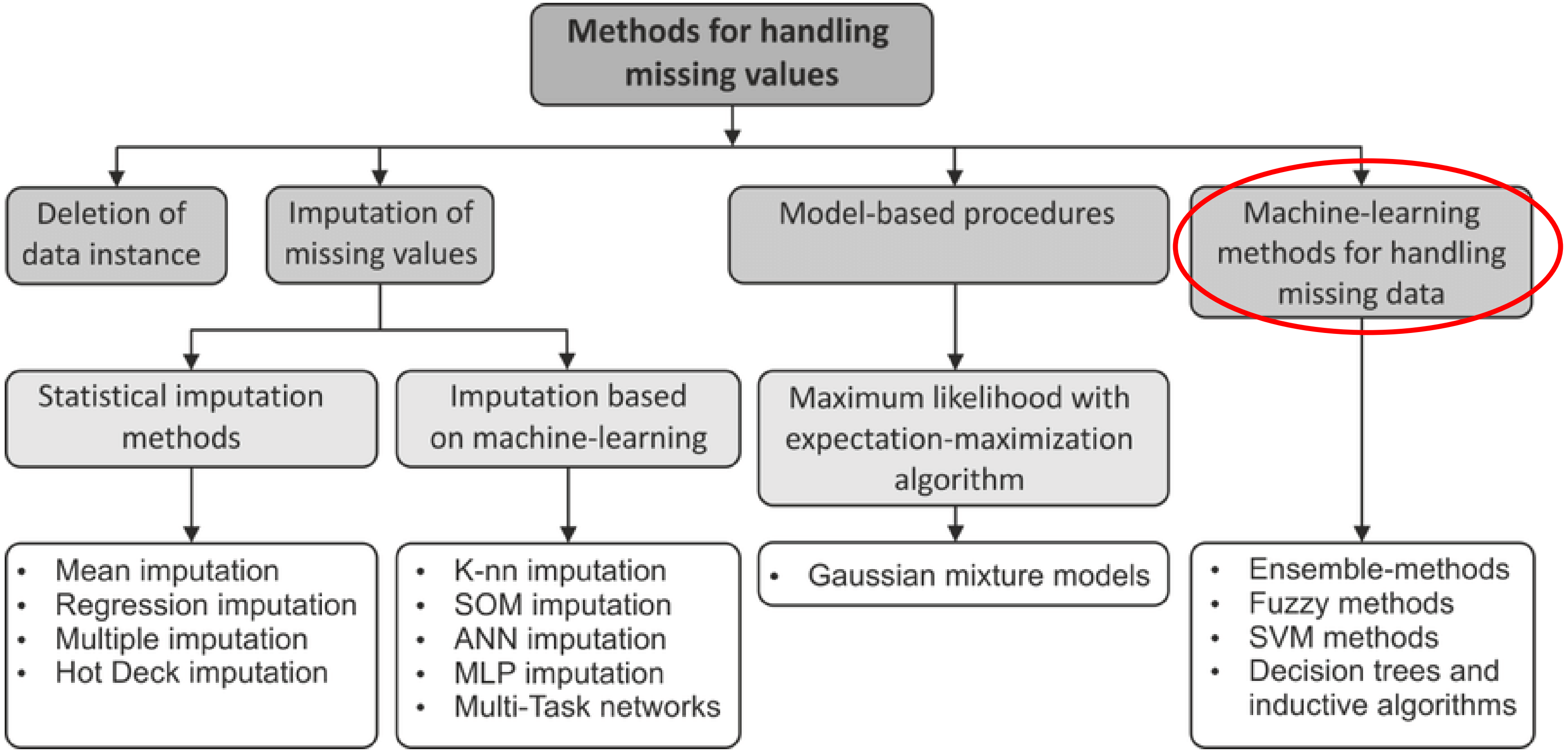
- Statistical methods based on constructing the likelihood function.
- Maximum likelihood estimation uses an iterative algorithm to estimate the parameters and the missing values based on the observed data and a set of assumptions.
- Mixture models allow for clustering by modeling the distribution of the observed data.



Machine-learning methods

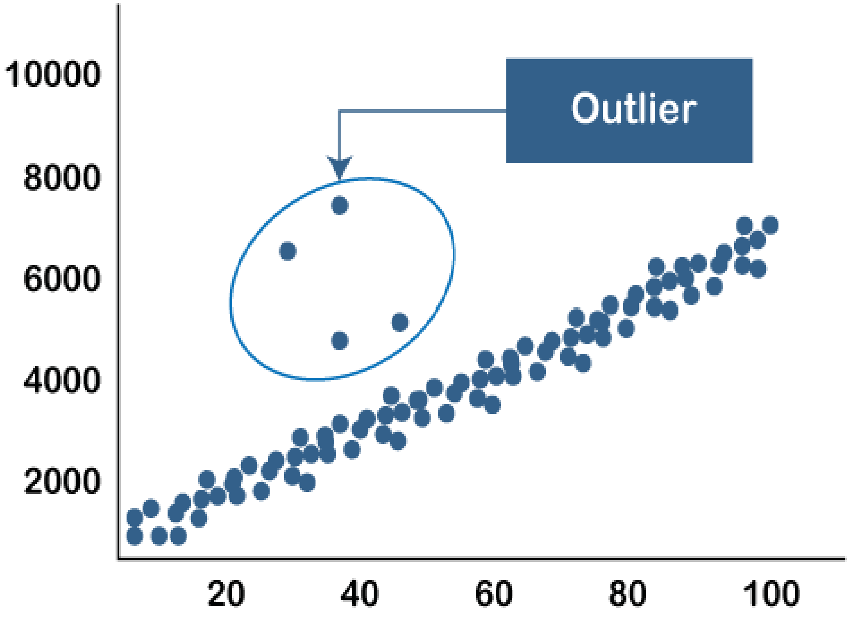
- The existence of missing values reduces the amount of knowledge learned by the machine learning models in the training stage thus affecting the classification accuracy, negatively.

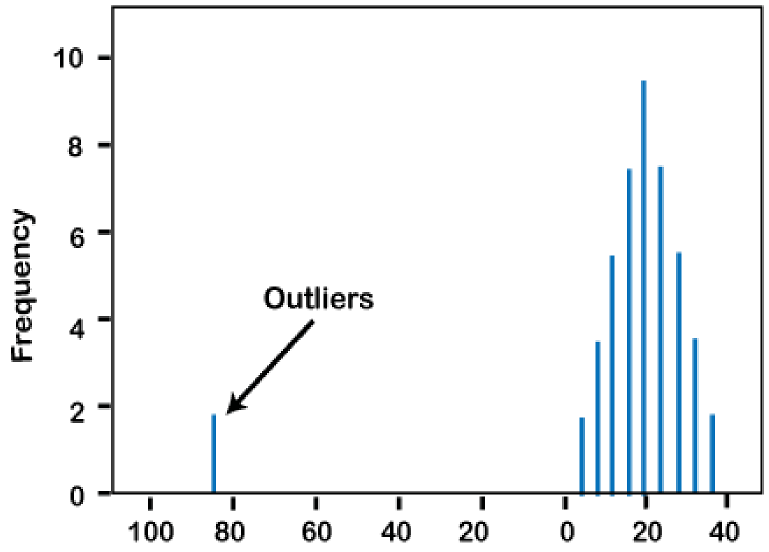
- Various algorithms such as Naive Bayes (NB), Support Vector Machine (SVM) and more can be used for handling the missing values.



Key steps involved in data preprocessing

- **Handling outliers**: *Identifying and handling outliers that could negatively impact the model's performance.*





Outlier Detection techniques: Numeric , Z-Score, DBSCAN, Isolation forest

Outliers vs. Noise

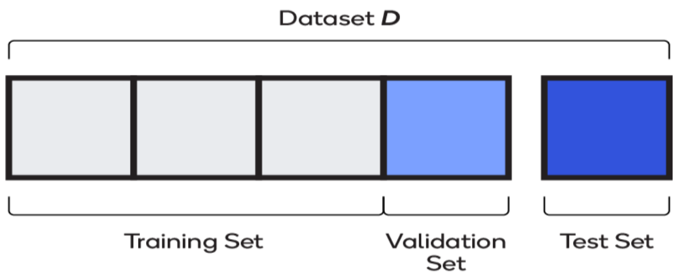
- Noise is a random error, but outlier is an observation point that is situated away from different observations.
- Noise should be removed for better outlier detection.

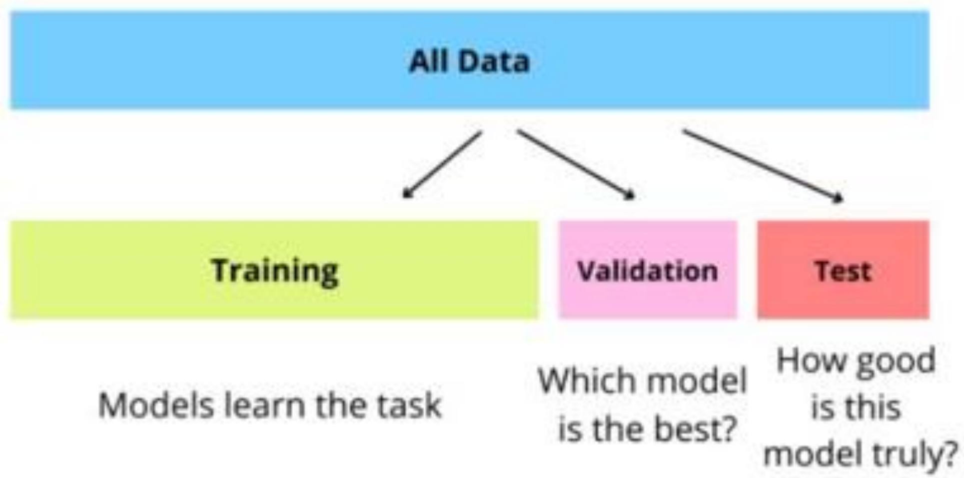
Key steps involved in data preprocessing

- **Data Splitting**: it is a critical step in the machine learning workflow, where the *available dataset is splitting into two or more subsets (e.g., training and testing sets) to evaluate the model's performance on unseen data.*

The typical data splitting ratios are:

- Training set: Around 60-80% of the data
- Validation set: Around 10-20% of the data
- Testing set: Around 10-20% of the data

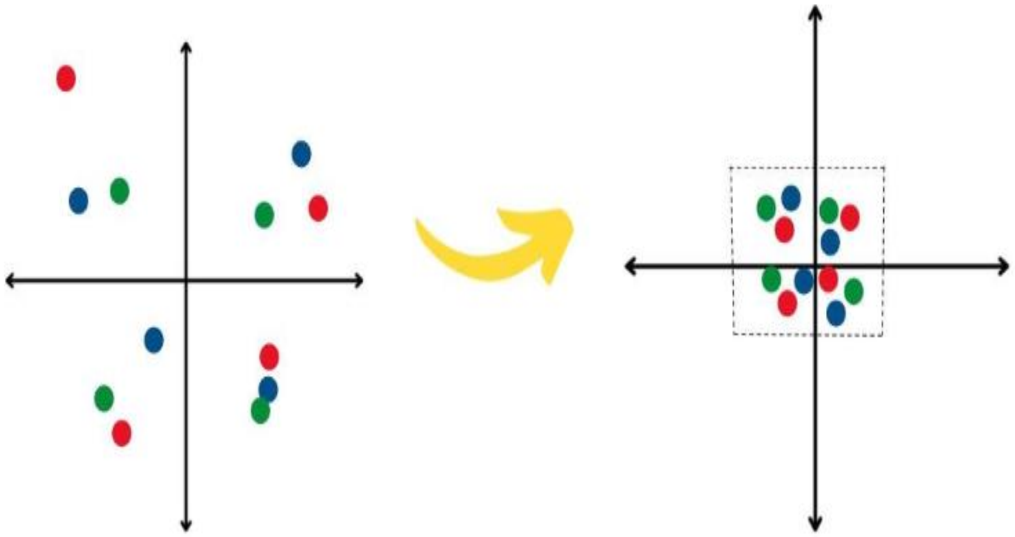


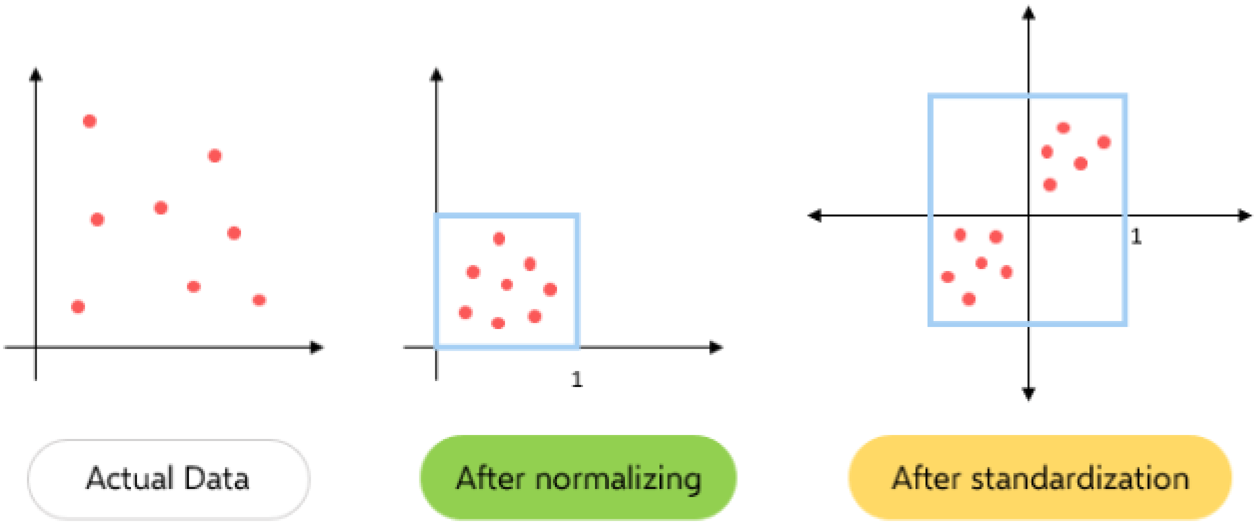


Key steps involved in data preprocessing

Feature Scaling: *Ensuring that all features have similar scales to avoid biases in certain machine learning algorithms. Common scaling techniques include* ***normalization (i.e.,*** Rescales data to a fixed range, usually [0,1]*) and* ***standardization (i.e.,*** Transform data to mean = 0, standard deviation = 1*).*

Because of features may have different ranges (e.g., age 0–100 vs. salary 0–100,000)





Common Normalization Techniques

Min–Max Normalization (Rescaling)	Decimal Scaling	Log Transformation	Unit Vector Normalization (L2 Normalization)
Scales values to a **fixed range** (commonly [0, 1] or [-1, 1]).	Moves the decimal point until all values fall in [-1, 1].	Applies log to compress large values and reduce skewness	Scales feature vectors to have unit length (norm = 1).
$x' = \frac{x - x_{min}}{x_{max} - x_{min}}$	$x' = \frac{x}{10^j}$ j is chosen so that $max(\lvert x' \rvert) < 1$	$x' = log(x + 1)$	$x' = x \,/\, \lvert\lvert x \rvert\rvert$
Ex: Age 10 in [0–100] → 0.1	Ex: If max = 987, divide by 10^3 = 1000 987 → 0.987	Ex: Sales data (highly skewed)	Common in text mining, NLP

Common Standardization Techniques

Z-Score Standardization	Robust Standardization (Median & IQR)
Converts data to have mean = 0 and std dev = 1. Implemented in sklearn as StandardScaler	Uses median and interquartile range (IQR) instead of mean/std
$x' = \frac{(x - \mu)}{\sigma}$	$x' = \frac{(x - median(x))}{IQR}$
Ex: Height 180 cm, μ=170, σ=10 → z=1	Ex: Income data

Key Summary

Method	Equation	Best for
Min–Max	$x' = \frac{(x - xmin)}{(xmax - xmin)}$	Distance-based models
Z–Score (Standardize)	$x' = (x - \mu)/\sigma$	Statistical models
Decimal Scaling	$x' = x / 10\hat{}j$	Simple quick scaling
Robust (IQR)	$x' = (x - median(x))/IQR$	Skewed data
Log Transform	$x' = \log(x + 1)$	Positive skewed data
L2 Normalization	$x' = x / \|\|x\|\|$	NLP, cosine similarity

Key steps involved in data preprocessing

Data Transformation: *This step involves converting the data into a consistent format. For example, converting categorical variables into numerical values through one-hot encoding or label encoding.*

Key steps involved in feature engineering

Feature Engineering: *Creating new features or transforming existing ones to enhance the model's predictive power. This might involve binning, polynomial features, log transformations, etc.*

Common techniques

- **Feature Creation**: This involves creating new features based on the existing ones or domain knowledge. **For example**, if the data contains a "date of birth" feature, one can create a new feature called "age" by subtracting the birth year from the current year.
- **Feature Encoding**: Converting categorical and textual data into numerical representations that can be effectively utilized by machine learning algorithms.
- **Feature Binning**: Grouping continuous values into bins to simplify the relationship between the feature and the target variable.

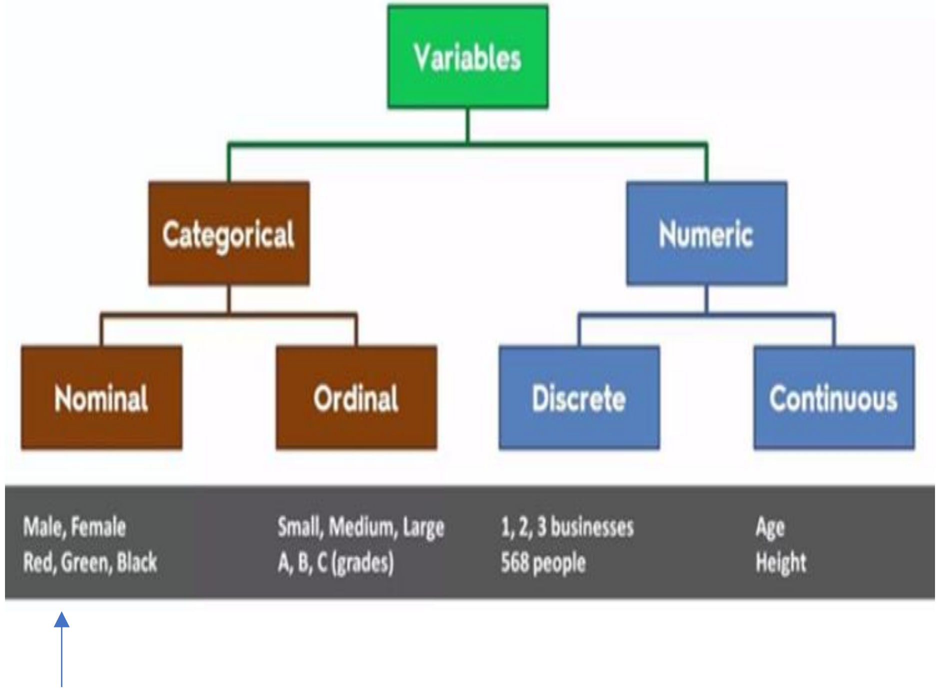
Feature Encoding

Feature Encoding: *It requires a deep understanding of the data and the problem domain. It often involves a trial-and-error process, where data scientists experiment with different feature engineering techniques to find the most informative representations of the data for the machine learning model.*

Some common types of feature encoding techniques:

- Label encoding
- Ordinal encoding
- One-hot encoding
- Dummy variable encoding
- Target encoding

Encoding is the process of converting data from one form to another. You don't change any information about the data, only representation is changing.



variables are just names and there is **no order or rank** to this variable's feature

Feature Encoding

Label encoding: *assigning a number value to each category to transform it to ordinal data. Each category is allocated a unique integer value.*

Encoding : [Red,Black,Yellow,Green] → [3 , 1 , 4 , 2]

Fruit	Color	Encode_color
Apple	Red	3
Graphs	Black	1
Banana	Yellow	4
strawberry	Red	3
pineapple	Yellow	4
kiwi	Green	2

Advantages:

- simple to do and keeps the order of the categories

Disadvantages:

- It may not always effectively reflect the underlying connections between the categories.
- It might cause problems when utilized with machine learning techniques.

Feature Encoding

Ordinal encoding: *assigning a number value to each variable where variables have some* ***order/rank (****maintaining their relationship)****.***

For example: Student's performance, Customer's review, Education of person, etc...

Original Encoding	Ordinal Encoding
Poor	1
Good	2
Very Good	3
Excellent	4

We have orders/ranks/sequences. We can assign ranks based on student's performance, based on feedback given by customers, based on the highest education of the person.

This approach is often used for ordinal categorical variables.

Feature Encoding

One-hot encoding: *Each category is represented by a binary vector of 0s and 1s. Each category gets its own binary feature, and only one of them is set to 1 at a time, indicating the presence of that category. Number of features = Number of unique categorical labels.*

color
red
green
blue
red

color_red	color_blue	color_green
1	0	0
0	0	1
0	1	0
1	0	0

Original Data

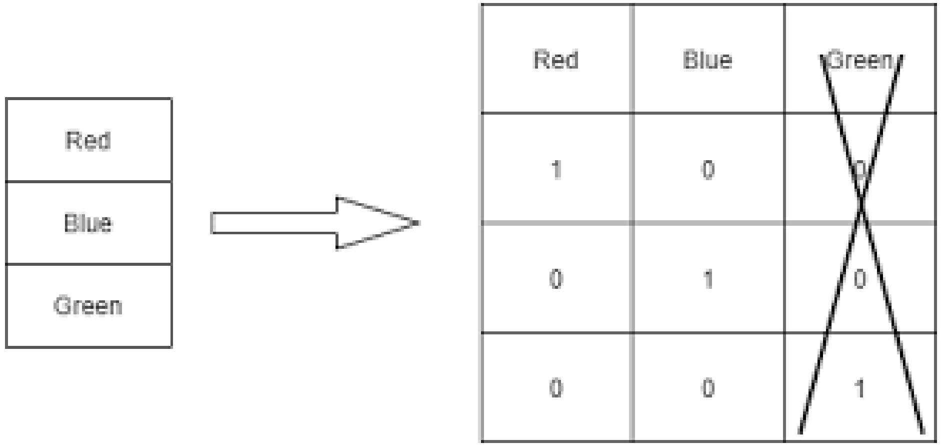
Team	Points
A	25
A	12
B	15
B	14
B	19
B	23
C	25
C	29

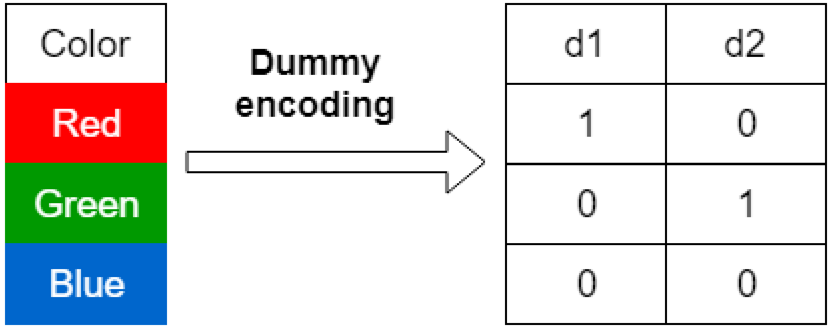
One-Hot Encoded Data

Team_A	Team_B	Team_C	Points
1	0	0	25
1	0	0	12
0	1	0	15
0	1	0	14
0	1	0	19
0	1	0	23
0	0	1	25
0	0	1	29

Feature Encoding

Dummy variable encoding: *Same as one-hot encoding but with one additional step. After one-hot encoding, we drop a feature randomly. Number of features = Number of unique categorical labels - 1*





Feature Encoding

Target encoding: *In this method, we calculate the mean of each categorical variable based on the output and then rank. It's particularly useful for classification tasks.*

Column	O/P
Btech	1
PHD	1
Master's	0
High School	0
PHD	1
Btech	0
Master's	0
High School	0
High School	1

Mean of Btech = 0.5

Mean of PHD = 1

Mean of Master's = 0

Mean of High School = 0.33

Column	Rank
Btech	2
PHD	1
Master's	4
High School	3
PHD	1
Btech	2
Master's	4
High School	3
High School	3

We can't do this with nominal as we don't know the order in case of nominal variables. To overcome this, use mean encoding.

Pincode	O/P
753001	1
753002	1
753003	0
753001	0
753004	1
753002	0
753002	1
753001	0
753003	1

Column	Mean
753001	0.33
753002	0.66
753003	0.5
753004	1

convert the categories into their mean values based on the output.

Categorial data encoding

① One-hot encoding

Color
Green
Red
Black
Orange

Green	Red	Black	Orange
1	0	0	0
0	1	0	0
0	0	1	0
0	0	0	1

② Dummy encoding

Green	Red	Black
1	0	0
0	1	0
0	0	1
0	0	0

Drop one feature (randomly)

③ Effect encoding

Green	Red	Black
1	0	0
0	1	0
0	0	1
-1	-1	-1

Alter row with all zeros to all -1

④ Label encoding

Color	Encoding
Green	1
Red	2
Black	3
Green	1

*unique labels

⑤ Ordinal encoding

Size	Encoding
S	2
XS	1
L	4
M	3

*data with order

⑥ Count encoding

Color	Encoding
Green	2
Red	1
Black	1
Green	2

*label based on frequency

⑦ Binary encoding

Color	Color_0	Color_1
Green	0	0
Red	0	1
Black	1	0
Orange	1	1

*binary labels

Feature Binning

Feature Binning: *Also known as **discretization** and used for the transformation of a continuous or numerical variable into a categorical/discrete feature. Binning of continuous variable introduces non-linearity and tends to improve the performance of the model.*

In other words, grouping continuous values into bins to simplify the relationship between the feature and the target variable.

Create bins and each bin allows a specific range of continuous numerical values. It prevents overfitting and increases the robustness of the model.

Feature binning techniques:

- Equal Width (or distance) Binning
- Equal Depth(or frequency) Binning
- Quantile Binning

Feature Binning

Equal Width Binning: *it divides the continuous variable into several categories having bins or ranges of the same width.*

- Let $\boldsymbol{x}$ be the number of categories and $\boldsymbol{max}$ and $\boldsymbol{min}$ be the maximum and minimum values in the concerned column. Then **width($\boldsymbol{w}$)** will be:-

$$w = \left\lfloor \frac{max - min}{x} \right\rfloor$$

and the categories will be:- $[min, min + w - 1], [min + w, min + 2 * w - 1], [min + 2 * w, min + 3 * w - 1] \cdots [min + (x - 1) * w, max]$

Example:- Let column X (AGE) be:-
X=[10,15,16,18,20,30,35,42,48,50,52,55] $min = 10, max = 55$, and let's say the number of categories "x" be 4. Then the width "w" will be : (55-10)/4 = 12.

The continuous data can be converted to categorical

AGE	AGE_bins
10	[10,21]
15	[10,21]
16	[10,21]
18	[10,21]
20	[10,21]
30	[22,33]
35	[34,45]
42	[34,45]
48	[46,55]
50	[46,55]
52	[46,55]
55	[46,55]

Feature Binning

Equal Depth Binning: *it divides the data into categories with approximately the same values.*

Let n be the number of data points and x be the number of categories required.

$$freq = \frac{n}{x}$$

Example:- Let column (AGE) be:-
n=12=[10,15,16,18,20,30,35,42,48,50,52,55] let's say the number of categories "x" be 4. Then the depth/frequency "$freq$" will be : 12/4 = 3.

The continuous data can be converted to categorical

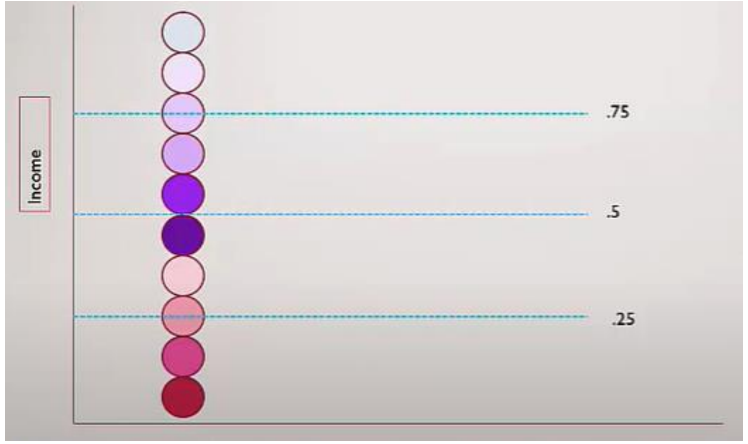
AGE	AGE_bins
10	[10,16]
15	[10,16]
16	[10,16]
18	[17,30]
20	[17,30]
30	[17,30]
35	[31,48]
42	[31,48]
48	[31,48]
50	[49,55]
52	[49,55]
55	[49,55]

Feature Binning

Quantile Binning: *is the process of assigning the same number of observations to each bin if the number of observations is evenly divisible by the number of bins.*

- The bin boundaries are ascertained by the values at specific percentiles (e.g., 25th, 50th, and 75th percentiles). Thus, **q-Quantiles** help in partitioning a numeric attribute into **q** equal partitions.

- **Ex:** **2-Quantile** known as the **median** which divides the data distribution into two equal bins, **4-Quantiles** known as the **quartiles** which divide the data into 4 equal bins and **10-Quantiles** also known as the **deciles** which create 10 equal width bins.



- Using fixed-width will not be effective if there are significant gaps in the range of the numerical feature, then there will be many empty bins with no data.

- Just like in frequency binning, quantiles are values that divide the data into equal portions. **The quantile method assigns values to bins based on percentile ranks.**

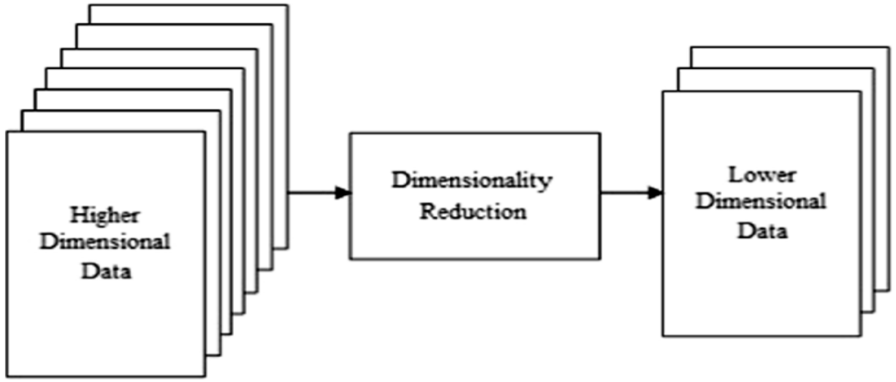
Key steps involved in feature engineering

Dimensionality reduction: *is a technique used to reduce the number of features in a dataset while retaining as much of the important information as possible. It used in ML and Data analysis.*

- In high-dimensional datasets, where the number of features is much larger than the number of samples, dimensionality reduction can help to overcome the "curse of dimensionality" and address various challenges, such as computational complexity, overfitting, and difficulty in visualizing and interpreting the data.

Goal

It represents the data in a lower-dimensional space while minimizing the loss of relevant information by reducing the number of features.



Benefits of Dimensionality Reduction

Improved Performance

Reducing the number of features can help machine learning algorithms generalize better and improve predictive performance by reducing the risk of overfitting.

Computation Efficiency

Lower-dimensional data simplifies computations, making it faster and more manageable for training machine learning models.

Visualization

Dimensionality reduction techniques often enable data visualization in 2D or 3D, making it easier to explore and interpret the data.

Data Compression

Reducing the number of features can lead to data compression, making it more efficient for storage and transmission.

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Dimensionality reduction techniques

Dimensionality Reduction

Technical | Approach

Feature Selection

Feature Selection is the process of selecting features/columns/variables, which contributes a major role to predict the independent variable or output in which we are interested during model building.

Filter
- Correlation
- Chi-Square test
- ANOVA
- Information Gain

Wrapper
- Forward Selection
- Backward Elimination
- Stepwise Selection

Embedded
- LASSO
- Elastic Net
- Ridge Regression

Feature Extraction

Feature Extraction process objective is to reduce the number of features and to extract/creating the relevant features and doing meaningful transformation prior to model building.

- Principal component analysis (PCA)
- Linear Discriminant Analysis (LDA)
- Etc.,

Key steps involved in feature engineering

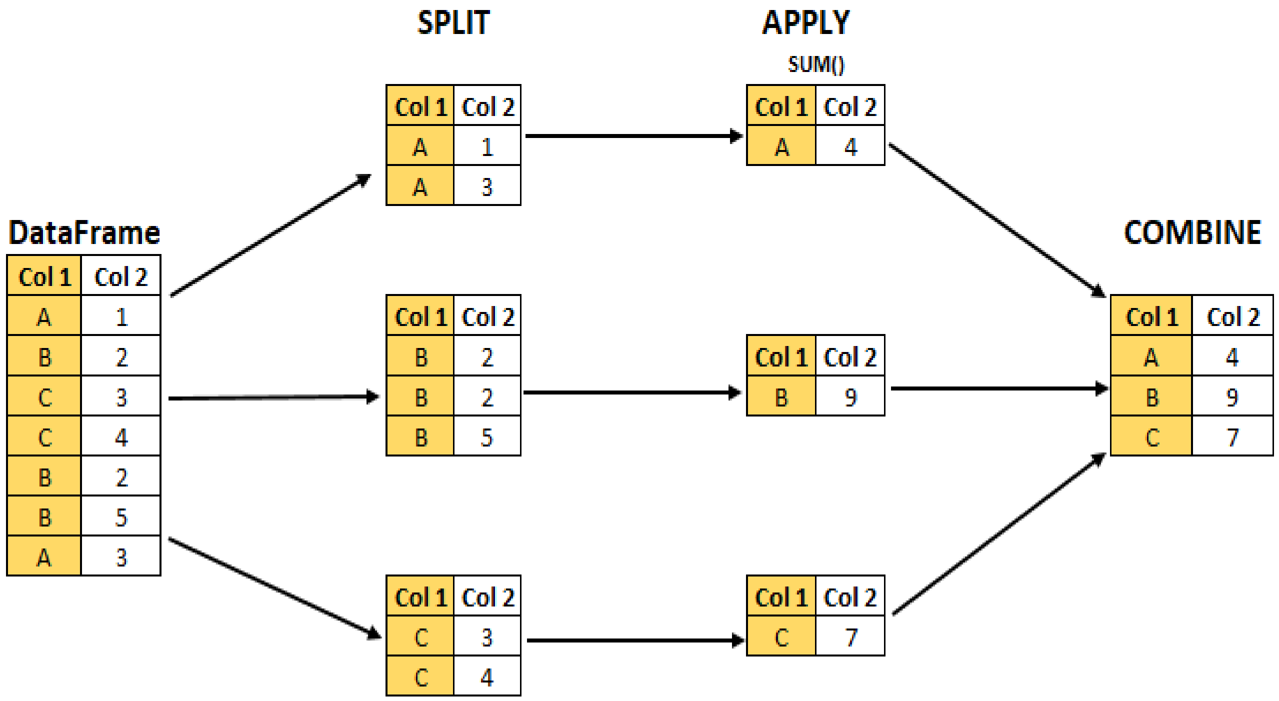
Grouping and aggregation: *Grouping data based on categorical variables and performing aggregate computations to gain insights.*

<u>**Ex:**</u> split-apply-combine operation

The ***split*** step involves breaking up and grouping a DataFrame depending on the value of the specified key.

The ***apply*** step involves computing some function, usually an aggregate, transformation, or filtering, within the individual groups.

The ***combine*** step merges the results of these operations into an output array.



Exploratory Data Analysis (EDA)

- Exploratory Data Analysis (EDA) refers to the process of visually and statistically examining the data to gain insights, discover patterns, and identify trends or anomalies.

- EDA plays a crucial role in the initial stages of data analysis and is essential for understanding the data and making informed decisions about data preprocessing and model selection.

Ex: Summary Statistics

Computing basic statistical measures such as mean, median, mode, Standard deviation, minimum, maximum, and quantiles to summarize the distribution of each numerical variable.

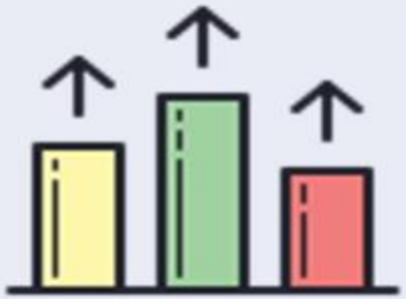
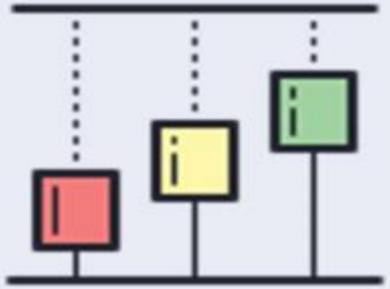
```
# Average petal length and width
iris_df.describe()
```

	sepal length (cm)	sepal width (cm)	petal length (cm)	petal width (cm)
count	150.000000	150.000000	150.000000	150.000000
mean	5.843333	3.057333	3.758000	1.199333
std	0.828066	0.435866	1.765298	0.762238
min	4.300000	2.000000	1.000000	0.100000
25%	5.100000	2.800000	1.600000	0.300000
50%	5.800000	3.000000	4.350000	1.300000
75%	6.400000	3.300000	5.100000	1.800000
max	7.900000	4.400000	6.900000	2.500000

Key steps involved in EDA

Data visualization: *Creating various plots and charts to visualize the data's characteristics.*

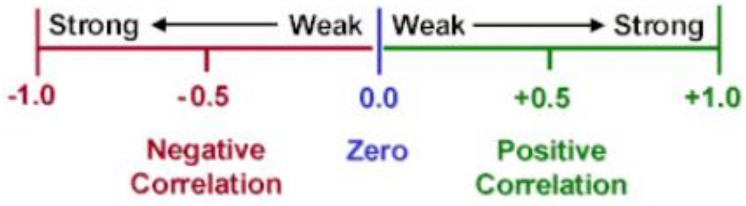
Common visualization techniques include:

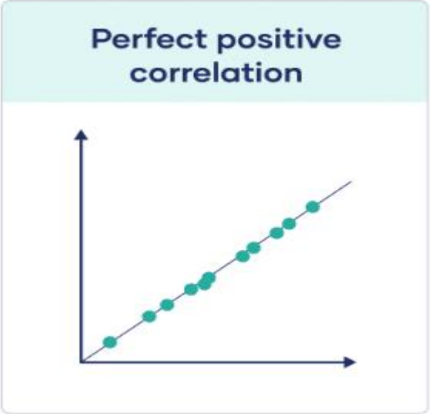
Heat Map	Pie Chart	Bar Chart	Histogram/Density Plot	Box Plot
Categorical	Categorical	Categorical	Numerical	Numerical
				
Visualizing correlations between numerical variables.	Visualizing part to whole .	Displaying the distribution of categorical variables.	Visualizing the distribution of numerical variables.	Displaying the spread and distribution of numerical data and detecting outliers.

Key steps involved in EDA

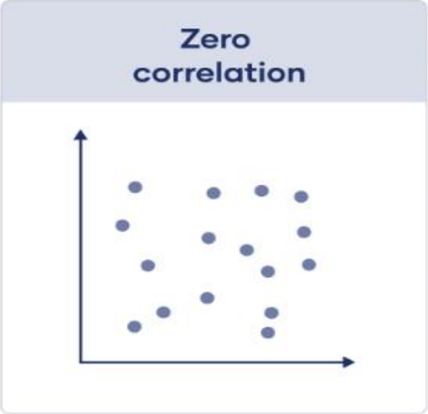
Correlation Analysis: *Calculating correlation coefficients between numerical variables to understand their interdependencies. In other words, it* ***reflects how similar the measurements of two or more variables are across a dataset.***

A **correlation coefficient** is a number between -1 and 1 that tells you the strength and direction of a relationship between variables.

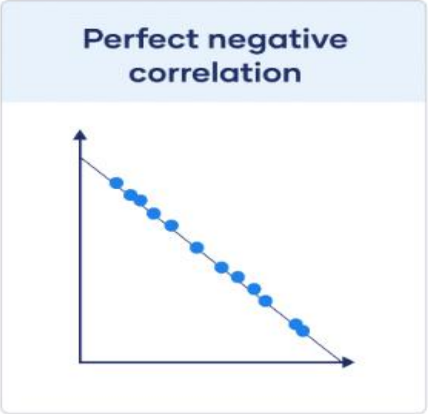




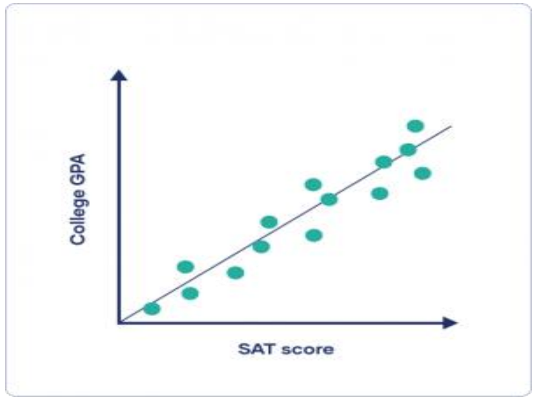
When one variable changes, the other variables change in the same direction.



There is no relationship between the variables.



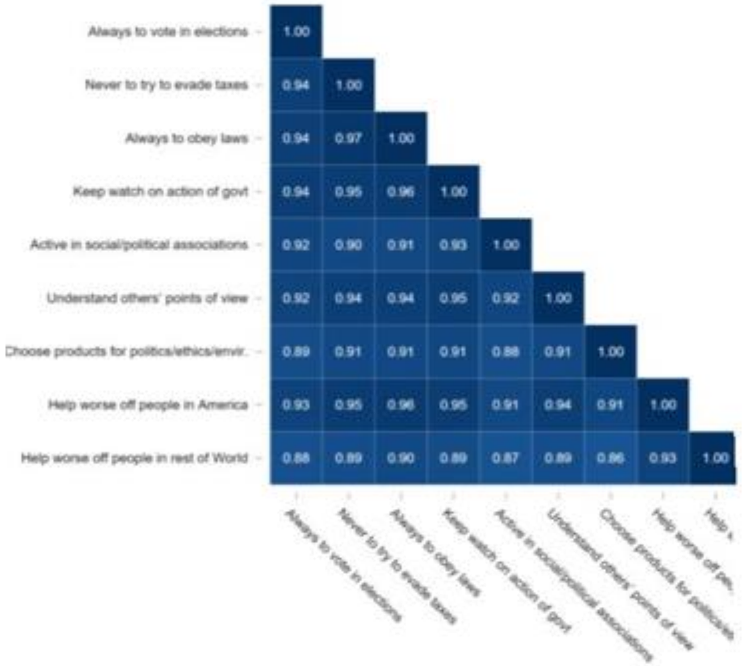
When one variable changes, the other variables change in the opposite direction.

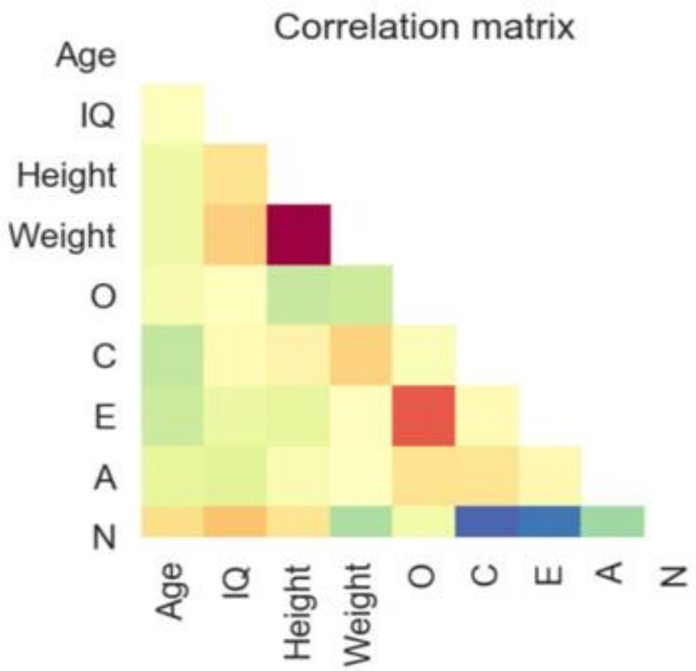


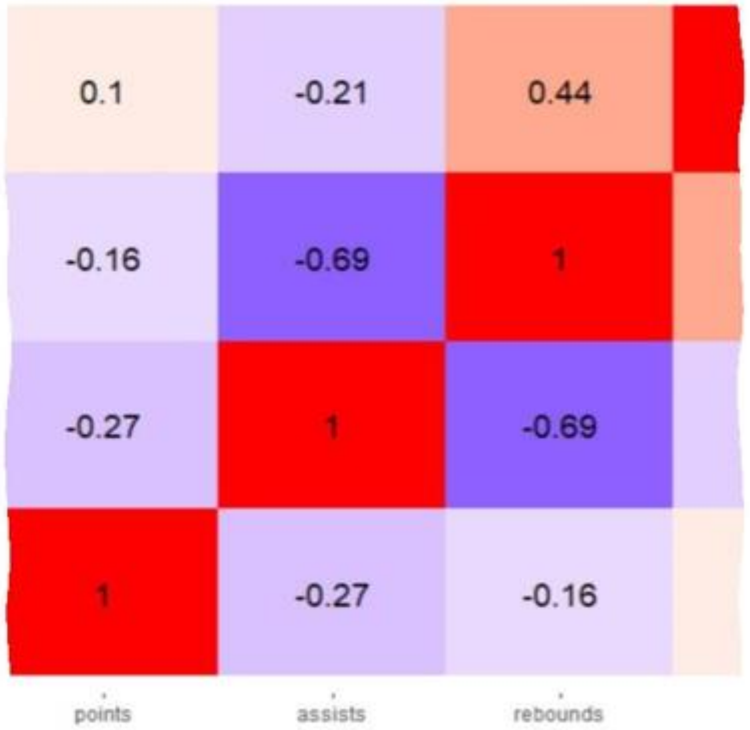
You predict that there's a positive correlation: higher SAT scores are associated with higher college GPAs while lower SAT scores are associated with lower college GPAs.

Key steps involved in EDA

Correlation matrix: *a square matrix showing the correlation coefficients between two variables. It is a useful tool for figuring out how different variables are related to each other.*







Model Selection

Model selection is the process of choosing the most suitable algorithm or model configuration that best captures the patterns in the data.

Purpose: To identify the model that provides the best performance without overfitting or underfitting.

Key Steps:

- Compare multiple candidate models (e.g., Linear Regression, Decision Tree, Random Forest, XGBoost).
- Train the model on a suitable dataset
- Use cross-validation to assess performance consistency.
- Evaluate models using metrics like accuracy, precision, recall, F1-score, RMSE, or R^2 depending on problem type.
- Consider complexity, interpretability, and computational cost.
- Tools & Libraries: Scikit-learn, TensorFlow, Keras, PyTorch, etc.

Model Training & Evaluation

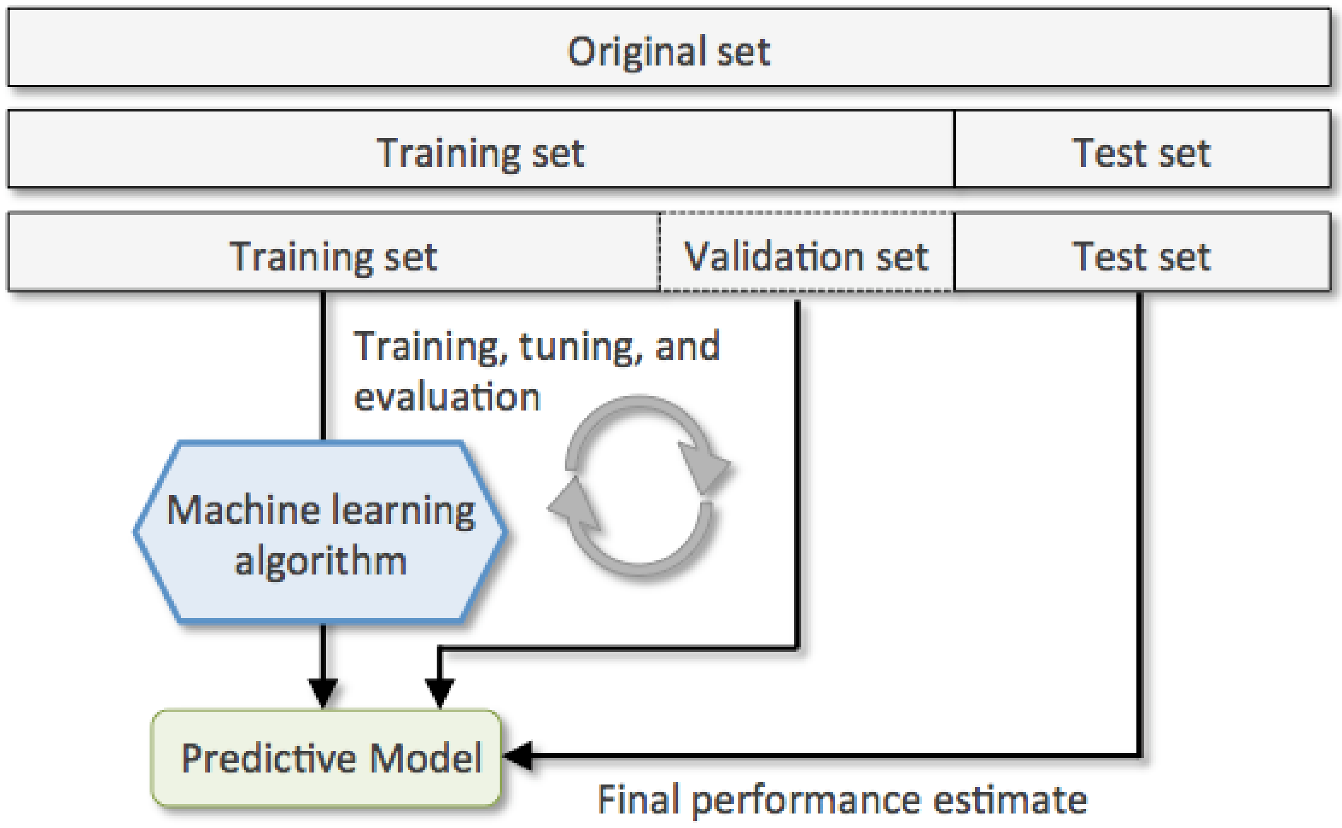
Model training involves teaching the model to learn patterns from data, while **evaluation** measures how well it performs on unseen data.

Purpose: To ensure that the model generalizes effectively and avoids overfitting or underfitting.

Key Steps:

- Split data into training, validation, and test sets.
- Train model using training data and tune hyperparameters on validation set.
- Evaluate model on test data to assess generalization.
- Use metrics such as accuracy, precision, recall, F1-score, confusion matrix, RMSE, or R^2.
- Perform error analysis to identify and improve weak areas.
- Tools & Frameworks: Scikit-learn, TensorFlow, Keras, PyTorch.

Model Training & Evaluation



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Model Deployment

Model deployment is the process of integrating a trained ML model into a production environment to generate predictions on new data.

Purpose: To operationalize the model so it provides value to end-users or systems.

Key Steps:

- Select deployment strategy: batch processing, real-time API, or edge device deployment.
- Package the model (e.g., Pickle, ONNX, TensorFlow SavedModel).
- Create APIs using frameworks such as Flask or FastAPI.
- Implement monitoring to track accuracy drift and data quality.
- Retrain and update the model periodically to maintain performance.
- Deployment Tools: AWS SageMaker, Azure ML, Google AI Platform, Docker, Kubernetes, Streamlit.

Test Your Knowledge

Q: Why is handling missing values important?

A: Missing data can bias results and reduce model accuracy; replacing or removing them ensures clean, reliable training data.

Q: What's the difference between outliers and noise?

A: *Noise* is random error in data; *outliers* are extreme values that differ significantly from the rest and can distort learning.

Q: What's the purpose of feature encoding?

A: To convert categorical variables into numerical form so ML algorithms can process them.

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THANK you